Instead of the number of alternatives *N*, scientists measure information in terms of a quantity called *S*, for 'Shannon information'. *S* is defined as the logarithm in base 2 of *N*: $S = \log_2 N$. The advantage of using the logarithm is that the unit of measurement $S = 1$ corresponds to $N = 2$ (because $1 = \log_2 2$), making the unit of information the minimum number of alternatives: the choice between *two* possibilities. This unit of measurement is called 'bit'. When I know at roulette that a red number has come up rather than a black, I have one bit of information; when I know that a *red, even* number has won, I have two bits of information; when an even red number 'manque' (eighteen or less, in roulette parlance) wins, I have three bits. Two bits of information correspond to four alternatives (red even, red uneven, black even, black uneven). Three bits of information correspond to eight alternatives. And so on.[fn48]

A key point is that information can be located somewhere. Imagine, for instance, that you have in your hand a ball which can be either black or white. Imagine that I also have a ball which can be either black or white. There are two possibilities on my part, and two on yours. The total number of possibilities is four (2 x 2): white-white; white-black; black-white and black-black. Now, suppose that for some reason we are certain that the two balls are opposite in colour (for instance, because we have taken the balls from a box that contained only one white and one black ball). The total number of alternatives is then *only 2* (white-black or black-white), even if the alternatives are still two on my part and two on yours. Note that, in this situation, something peculiar happens: if you look at your ball, then *you know the colour of mine*. In this case, we say that the colours of the two balls are correlated, that is to say, linked to one another. We say that my ball 'has information' about yours (as well as vice versa).

If you think about it, this is precisely what happens in life when we communicate: for example, when I phone you, I know that the phone causes the sounds on your side to be dependent on the sounds on mine. The sounds on both sides are linked, like the colour of the balls.

The example is not chosen at random: Shannon, who invented the theory of information, worked for a telephone company, and was looking for a way to measure accurately how much a telephone line could 'carry'. But *what* does a telephone line carry? It carries information. It carries the capacity to distinguish between alternatives. For this reason, Shannon defined information.

Why is the notion of information useful, perhaps even fundamental, to understanding the world? For a subtle reason: because it measures the ability of one physical system to communicate with another physical system.